

Q1. Create a dataframe containing information about different products such as Price, Rating, and Number of Sales. Generate a Dendrogram to group similar products based on their characteristics.

[10 Marks]

Q2. Create a suitable dataset named Loan_application.csv containing at least 10 records with the attributes Application_ID, Age, Income, Credit_Score, Loan_Amount, Employment_Years, Debt_Ratio, and Loan_Status, where Loan_Status represents whether the loan application is Approved or Rejected.

Write a Python program to perform the following data reduction techniques:

- a) Feature Selection: Identify the important features for predicting Loan_Status.
- b) Feature Extraction: Reduce the number of numerical attributes using a suitable technique such as Principal Component Analysis (PCA).

[20 Marks]

OR

Q2. Using the built-in Iris dataset, apply K-Means clustering with $K = 3$. Plot the datapoints, visualize the clusters with different colors, and display the cluster centroids.

[20 Marks]

Q3. Viva

[5 Mark]